CS 0007: Introduction to Java Lecture 1

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Not Your Usual Best Buy Pitch for a Computer

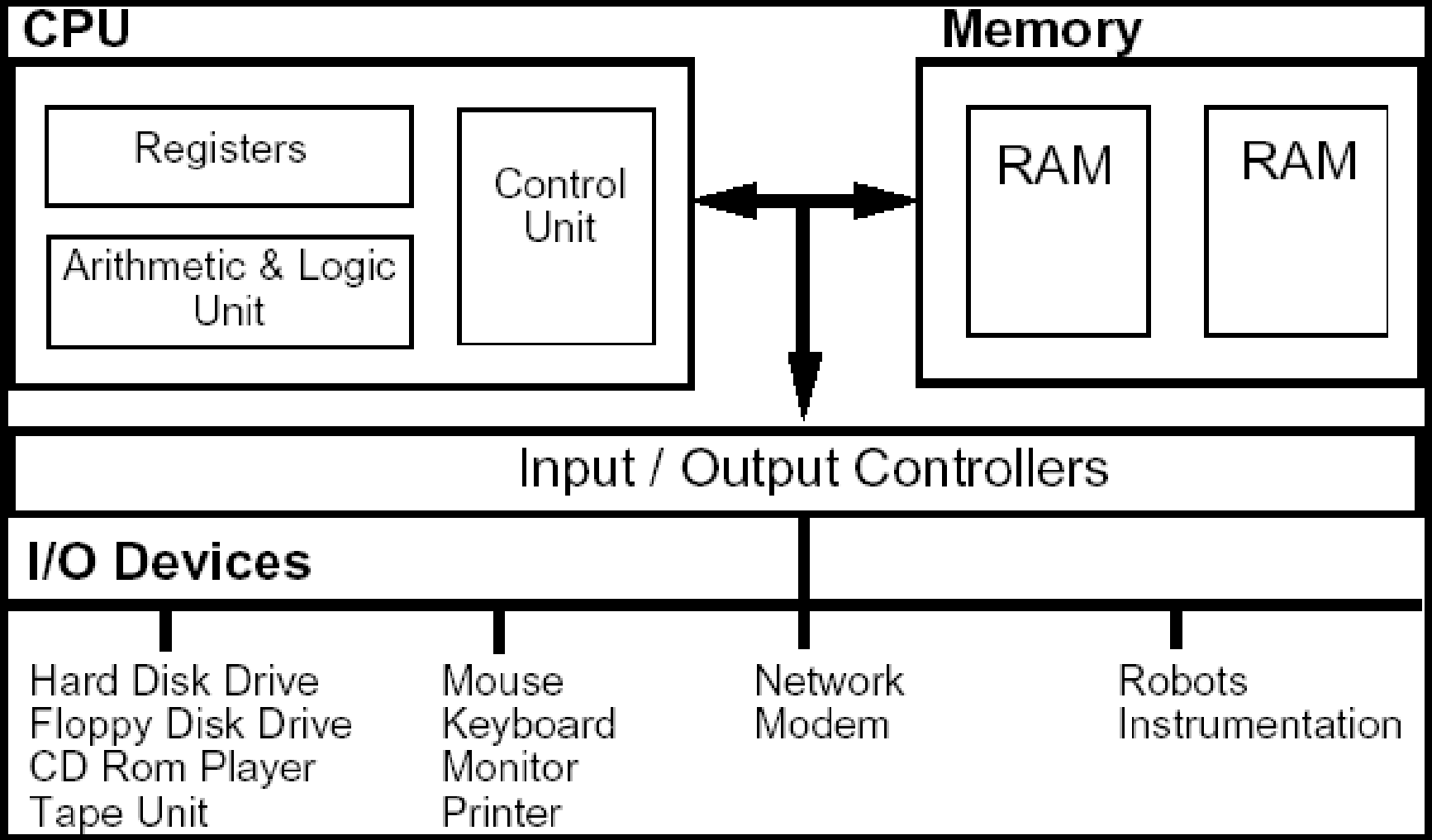
AND JAVA IS NOT JUST AN INDONESIAN ISLAND OR A TYPE OF COFFEE

What Happens in that Magical Box?



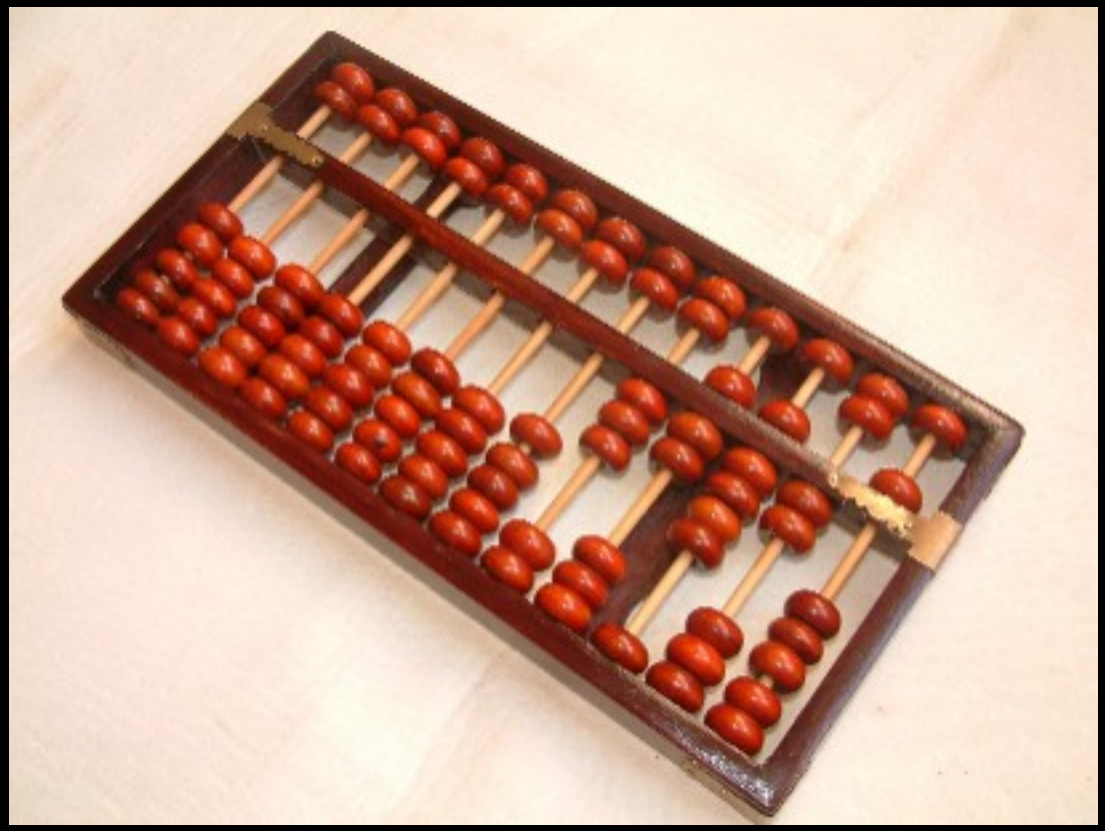
Source: http://ak1.ostkcdn.com/images/products/77/822/L13881092.jpg

How a Computer is Organized



Source: https://www.doc.ic.ac.uk/~eedwards/compsys/overall.gif

History of Computing



Source: https://upload.wikimedia.org/wikipedia/commons/e/ea/Boulier1.JPG

History of Computing

??? – Math invented

~2400 BC – Abacus probably invented in Babylon

??? – More math invented

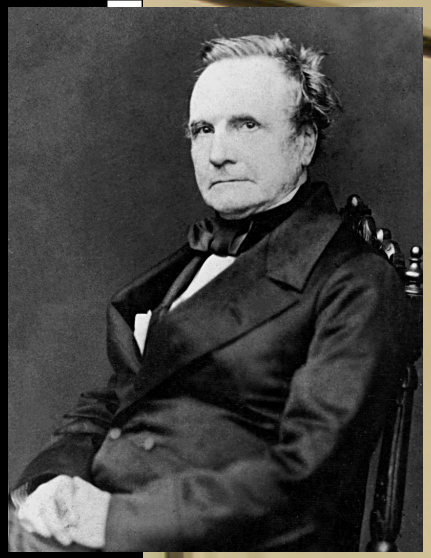
1600s – John Napier discovers logarithms and fast log computation

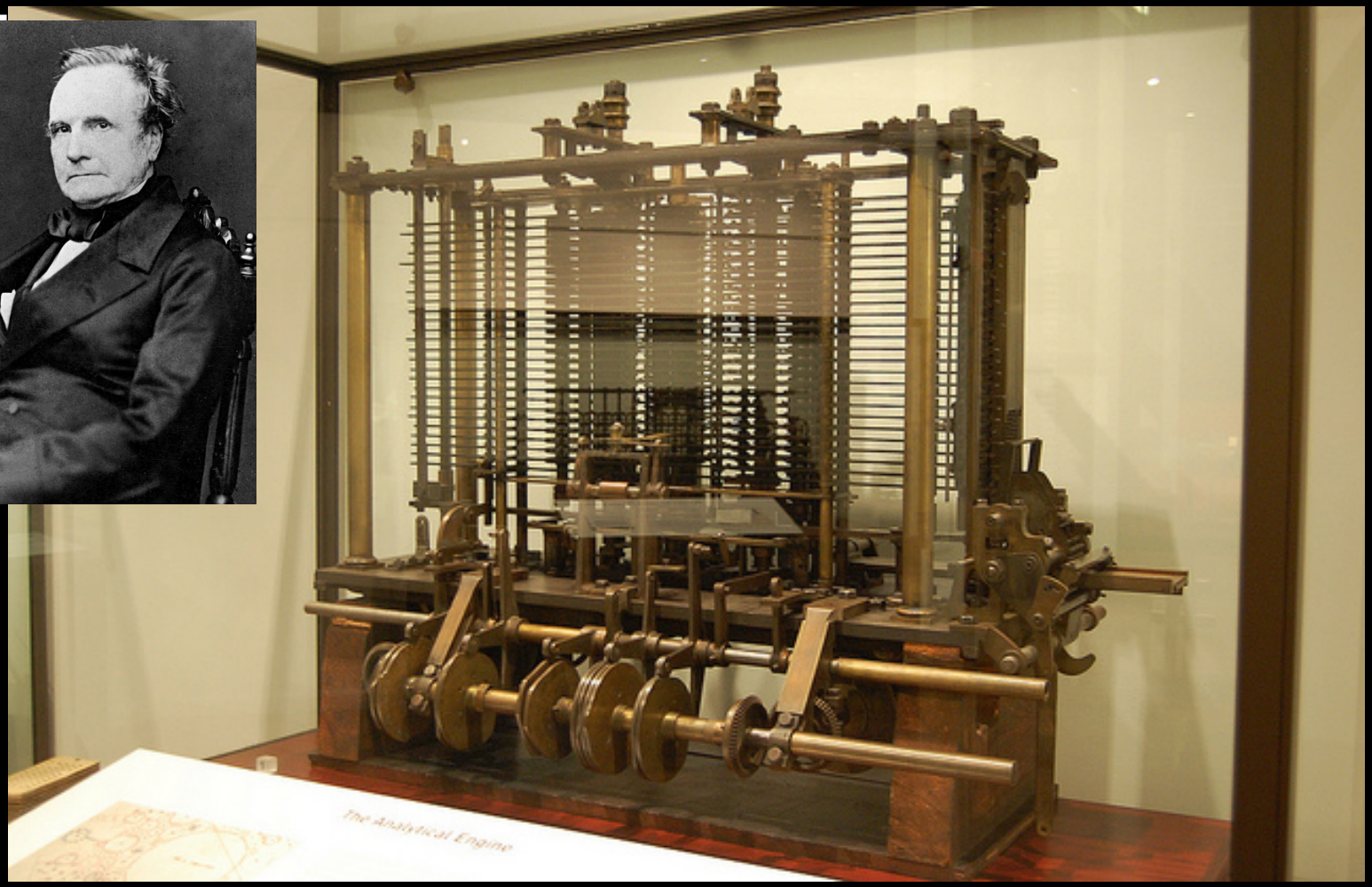
1786 – Johann Müller theorizes a "Difference Engine"

1822 – Charles Babbage ("father of the computer") secures funding to build a Difference Engine (never built)

History of Computing

1837 – Charles Babbage designs "Analytical





1841 Prototype on display at The Science Museum in London
(Source: https://s3.yimg.com/te/8543/8649286817_a3042f2c29_z.jpg)

Analytical Engine

- Babbage's Engine is considered the first general-purpose computer
- Could handle basic math operations, logic, loops, conditional branching, and had some primitive memory
- Powered by a hand-crank
- The full-scale version was never built, due to inadequate funding

Analytical Engine

- Accepted two kinds of punch cards:
  - Top: "Variable Cards" (Data)
  - Bottom: "Operational Cards" (Instructions)
- Punch cards were not new (used to direct mechanical looms at the time)



Source: https://upload.wikimedia.org/wikipedia/commons/0/0d/PunchedCardsAnalyticalEngine.jpg

Ada Lovelace

- In 1842, she recognized the usage of multiple, non-direct, complex operations and memory could generate Bernoulli numbers using Babbage's Analytical Engine.
- This is the first computer program.



Source: https://upload.wikimedia.org/wikipedia/commons/a/a4/Ada_Lovelace_portrait.jp

Side note: Yes there is a fan-fiction ship musical for Babbage and Lovelace.

History of Computing

1885 – AT&T (the company behind the operating system Unix and the programming language C) founded

1906 – Xerox (the company behind desktops and the computer mouse) founded

1907 – Vacuum tube invented

1911 – IBM (the company behind hard disks and floppy disks) founded

1931 – Charles Wynn-Williams published first usage of electronics in computation

History of Computing

1938 – Claude Shannon published first usage of electronics in Boolean algebra computation

1946 – Alan Turing publishes a paper on programs stored and read from tape

– The ENIAC, the first electronic general-purpose computer, was developed

1947 – Transistor invented

– Cathode Ray Tube (CRT) Random Access Memory (RAM) invented

History of Computing

1952 – Grace Hopper invents first compiler
1956 – Hard disk invented
1957 – FORTRAN invented
1958 – Integrated circuit (chip) invented
– LISP invented
1960 – COBOL invented
1962 – First computer game created
1963 – Computer mouse invented

History of Computing

1964 – BASIC invented

1965 – First supercomputer developed

– Gordon Moore publishes his famous "Moore's Law"

1968 – Moore founds Intel (the company behind CPUs and other chips)

1969 – ARPANET (predecessor to the Internet) developed

– UNIX developed

1970 – Integrated Circuit Dynamic-RAM developed

– Pascal invented

History of Computing

1971 – Intel develops first commercially available microprocessor (4004)

– Famous "P vs. NP" problem stated by Kurt Gödel

– Floppy disks invented

1972 – C invented

1975 – First personal computer sold (Altair 8800)

– Microsoft founded

1976 – Apple founded

– LISP invented

History of Computing

1978 – SQL invented

1982 – Sun Microsystems (company behind Java) founded

1984 – C++ invented

1985 – CD-ROM developed

1989 – World Wide Web invented

1991 – Linux invented

1993 – Concept of an Internet search engine developed

History of Computing

1994 – First smartphone developed

1995 – USB invented

– DVD-ROM invented

– Java invented

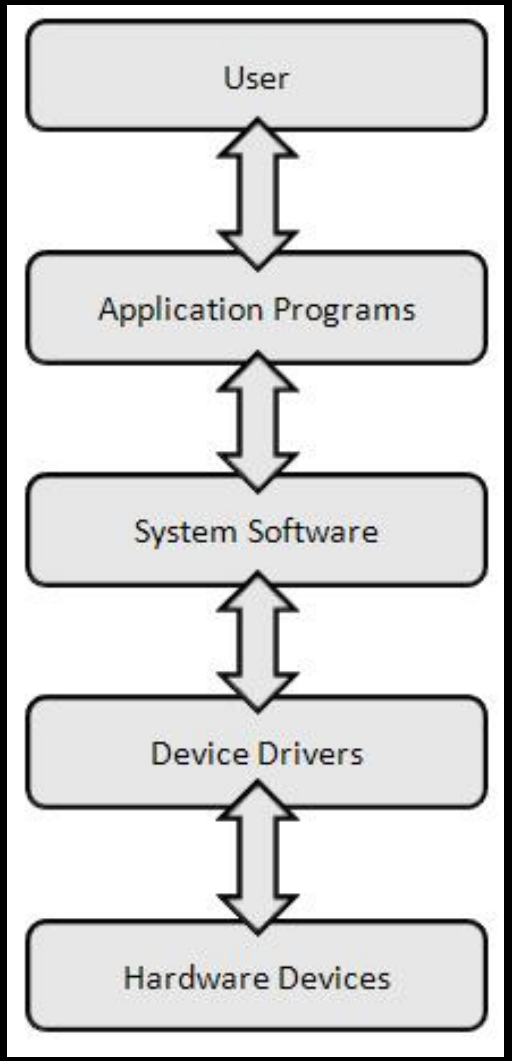
Java Today

- Oracle acquired Sun Microsystems in 2010.
- One of the most widely used languages, primarily found in business enterprises, web applications, or mobile phones (i.e. Android).
- Current version: Java SE 8u102

Why Learn Java?

- Widely used
- Constantly updating and receiving support
- Easily understood due to standardized and consistent syntax
- Forces code readability
- Virtual Machine isolates running code

Software Layers



- Applications: Software that interfaces with the user
- System Software: Operating System (e.g. Windows, Mac OSX, Linux)
- Device Drivers: Software to talk to non-computer devices (e.g. printers, webcams, etc.)

Source: http://cs.sru.edu/~mullins/cpsc100book/images/softwareLayers.jpg